

Reconstruction of Absorption Spectra of Healthy Renal Tissue and Chromophobe Renal Cell Carcinoma Using Non-Negative Least Squares Fitting

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Abstract—Chromophobe renal cell carcinoma (ChRCC) is a subtype of renal cell carcinoma whose accurate identification remains important for appropriate clinical management. Optical spectroscopy has emerged as a promising approach for tissue characterization due to its ability to provide biochemical information in a rapid and minimally invasive manner. In this work, the absorption spectra of healthy renal tissue and ChRCC were reconstructed from the absorption spectra of their principal biological components using a non-negative least squares fitting approach. Absorption coefficient spectra were obtained from total transmittance and total reflectance measurements acquired in the 200–1000 nm spectral range. Reference absorption spectra of DNA, proteins, oxygenated hemoglobin, deoxygenated hemoglobin, water, lipids, melanin, and lipofuscin were used to reconstruct the experimental tissue spectra and estimate chromophore concentrations. The reconstructed spectra showed excellent agreement with the experimental data, with mean relative differences of 1.04% for healthy tissue and 0.89% for ChRCC tissue. Quantitative analysis revealed increased DNA and protein concentrations in ChRCC, consistent with the enhanced cellular proliferation and metabolic activity associated with tumour development. The obtained results demonstrate that broadband absorption spectroscopy combined with non-negative least squares spectral reconstruction enables accurate characterization of renal tissue composition and represents a promising approach for future optical diagnostic applications in renal cancer.

Keywords—Chromophobe renal cell carcinoma, renal tissue, absorption spectroscopy, optical properties, chromophores, spectral reconstruction, non-negative least squares, biophotonics, tissue characterization.

I. INTRODUCTION

The kidneys are vital paired organs located retroperitoneally, whose primary functions include the filtration of blood and the formation of urine for the elimination of metabolic waste products, the regulation of acid-base balance, the maintenance of extracellular fluid homeostasis, electrolyte regulation, and the secretion of hormones involved in blood pressure control and erythropoiesis via the renin-angiotensin-aldosterone and erythropoietin axes [1]. Given the central role of the kidneys in systemic homeostasis, diseases affecting these organs carry significant clinical consequences. Among renal malignancies, renal cell carcinoma (RCC) constitutes the most frequent

kidney cancer in the adult population and one of the most lethal malignant tumours worldwide [2]–[4].

RCC comprises a heterogeneous group of malignancies arising from the renal tubular epithelium, each characterised by distinct morphological, molecular, and clinical features [5]. Clear cell RCC (ccRCC) is the most prevalent subtype, accounting for approximately 75–80% of all cases [2]. The remaining 20–25% of cases are classified under non-clear cell or variant histology RCC, including papillary RCC (pRCC) and chromophobe RCC (ChRCC), among others [5], [6]. ChRCC is the third most common subtype, representing approximately 5% of all RCC cases, and arises from the distal tubules and collecting ducts of the nephron [6], [7]. Although ChRCC is generally associated with a more favourable prognosis than ccRCC, accurate subtype identification remains essential for appropriate clinical management and treatment planning [2], [7].

The clinical management and prognosis of RCC are strongly influenced by the stage at diagnosis. A major challenge is that most patients remain asymptomatic during the early stages of the disease, resulting in many tumours being detected incidentally during abdominal imaging performed for unrelated clinical indications [2], [3]. Current diagnostic evaluation relies primarily on contrast-enhanced computed tomography (CT), magnetic resonance imaging (MRI), and renal ultrasound, while histopathological assessment of biopsy or surgical specimens remains the gold standard for definitive diagnosis [3], [7]. However, these approaches present important limitations. Imaging modalities are often unable to reliably differentiate RCC subtypes, as the radiological appearance of ChRCC may overlap considerably with that of ccRCC and renal oncocytoma [7]. In addition, renal mass biopsy is subject to sampling errors, delayed pathological assessment, and the potential need for repeated procedures to ensure diagnostic accuracy. These limitations highlight the need for complementary diagnostic approaches capable of providing rapid, accurate, and minimally invasive tissue characterization [8].

Biophotonics, the interdisciplinary field that studies the interaction between light and biological matter, has emerged as a powerful tool for biomedical diagnostics, imaging, and therapy. Advances in optical technologies have enabled the development of techniques capable of characterizing biological tissues in a rapid and minimally invasive manner [9]. Among these techniques, diffuse reflectance spectroscopy (DRS), fluorescence spectroscopy, Raman spectroscopy, and optical coherence tomography (OCT) have demonstrated potential for tissue characterization through the assessment of tissue optical properties [8], [10], [11].

In reality, DRS has attracted particular interest, since it is a label-free technique capable of probing tissue composition through the analysis of light that has undergone multiple scattering and absorption events within biological tissues [10]. Optical spectroscopy offers real-time, minimally invasive biochemical tissue characterization without the need for exogenous contrast agents [8]–[10]. In this context, the optical properties of biological tissues play a fundamental role, as they determine how light propagates, scatters, and is absorbed within tissue [10], [11].

The interaction of light with biological tissue is governed by its optical properties, principally the absorption coefficient (μ_a), the scattering coefficient (μ_s), the anisotropy factor (g), and the refractive index (n) [12]. Among these properties, the spectral absorption coefficient, $\mu_a(\lambda)$, is of particular interest because it provides direct information regarding the chromophores present in a tissue and their relative contributions [12]. The absorption coefficient at a given wavelength λ can be estimated from spectral transmittance and reflectance measurements of a tissue sample of known thickness d according to [12]:

$$\mu_a(\lambda) = \frac{[1 - (T_t(\lambda) + R_t(\lambda))]}{d}, \quad (1)$$

where $T_t(\lambda)$ and $R_t(\lambda)$ correspond to the total transmittance and total reflectance spectra, respectively [12]. Since biological tissues are heterogeneous media composed of multiple absorbing constituents, their broadband absorption spectra can be represented as a weighted sum of the absorption spectra of the individual tissue components [13], [14]:

$$\mu_a(\lambda) = \sum_i w_i \cdot \mu_{a,i}(\lambda), \quad (2)$$

where w_i represents the contribution of the i -th component and $\mu_{a,i}(\lambda)$ corresponds to its absorption spectrum. This formulation enables the identification and quantification of tissue chromophores from measured absorption spectra [12].

Alterations in the concentration of these chromophores are closely associated with the structural and metabolic changes that occur during tumour development and progression. Tumour tissues frequently exhibit changes in nucleic acid, protein, lipid, and haemoglobin content, which are reflected in their optical absorption characteristics [8], [15], [16]. These compositional differences give rise to measurable variations in the absorption spectra of healthy and pathological tissues,

providing the biological basis for spectroscopic tissue discrimination [8], [12]. Therefore, the quantitative estimation of chromophore contributions from measured tissue absorption spectra may provide biologically meaningful information for objective tissue characterization and disease diagnosis [12].

While the broadband absorption spectrum measured from biological tissue represents the combined contributions of multiple chromophores, the decomposition of this composite signal into its individual constituents remains a challenging task. The substantial spectral overlap between biological chromophores across the ultraviolet, visible, and near-infrared regions further complicates the identification of individual absorption contributions [12]. Among the available spectral decomposition approaches, the least squares method provides a simple and effective framework for estimating the contribution of individual chromophores to a measured tissue absorption spectrum [17]. In matrix form, the least squares solution can be expressed as [18]:

$$\mathbf{b} = (X^T X)^{-1} X^T \mathbf{y}, \quad (3)$$

where $\mathbf{b}$ contains the estimated chromophore contributions, X contains the reference absorption spectra of the tissue components, and $\mathbf{y}$ corresponds to the measured tissue absorption spectrum [17]. By minimising the difference between the experimental spectrum and a weighted combination of reference spectra, this approach enables quantitative estimation of chromophore contributions and provides biologically meaningful information regarding tissue composition [12], [17].

Despite the recognised diagnostic value of optical spectroscopy for renal tumour characterisation, quantitative reconstruction of the absorption spectra of healthy renal tissue and ChRCC from the spectra of their constituent chromophores remains insufficiently explored. By modelling tissue absorption as a weighted combination of the absorption spectra of its principal biological components, it becomes possible to estimate the contribution of water, haemoglobin, nucleic acids, proteins, lipids, and pigments to the overall tissue absorption spectrum [12]. Such an approach may provide quantitative information regarding tissue composition and facilitate the identification of spectral biomarkers associated with pathological changes. Therefore, the objective of this work is to reconstruct the absorption spectra of healthy human renal tissue and ChRCC using the reference absorption spectra of their principal biological components and a non-negative least squares fitting approach, enabling a quantitative and component-resolved characterization of renal tissue optical properties in health and disease.

Section II describes the materials and methods employed for the reconstruction of the absorption spectra of healthy renal tissue and ChRCC, while section III presents and discusses the obtained results.

II. MATERIALS AND METHODS

This chapter describes the materials and methods used to reconstruct the absorption spectra of healthy renal tissue and cancerous renal tissue. First, the experimental procedures used

to acquire the spectral data are presented, followed by the computational methodology employed to calculate the absorption spectra, perform spectral reconstruction, and estimate the concentrations of the tissue's various biological components.

A. Sample preparation and spectral acquisition

For this study, tissue samples obtained from renal resections were used, sourced from the Portuguese Institute of Oncology in Porto (IPO-Porto), Portugal. All patients provided informed consent for the use of the collected samples in research activities, and the study was approved by the Ethics Committee of IPO-Porto. The samples, which contained both healthy tissue and tumor tissue, underwent histopathological analysis, which confirmed the presence of ChRCC in the selected tumor regions. Sample preparation was performed using a cryostat-microtome (CM1850 UV, Leica™, Wetzlar, Germany), allowing for the collection of 10 samples of healthy renal tissue and 10 samples of tumor tissue (ChRCC). All samples were prepared with an approximately circular geometry, having a diameter of about 1 cm and a uniform thickness of 0.5 mm, and were subsequently used for spectral acquisition. Prior to spectral acquisition, the samples were kept in saline solution for 10 min to ensure hydration conditions similar to those found in biological tissue *in vivo*. The main steps of this process are illustrated in Figure 1.

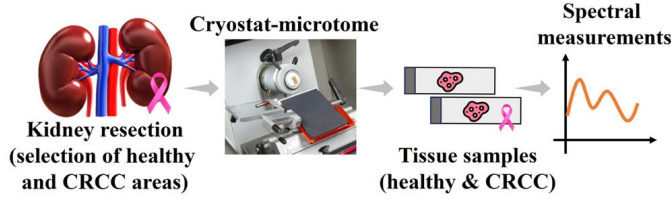


Fig. 1: Steps for collecting and preparing samples of healthy renal tissue and ChRCC used in spectral characterization.

B. Spectral Data

The spectral data required to reconstruct the tissue spectra were obtained using an experimental setup consisting of a xenon lamp (Avalight-XE-HP, Avantes™), optical fibers, an integrating sphere (AvaSphere-50, Avantes™), and a spectrophotometer (AvaSpec-2048-USB2, Avantes™). Two distinct experimental configurations, shown in Figure 2, were used to obtain the total transmittance (T_t) and total reflectance (R_t) spectra [19]. In the configuration designed for measuring total transmittance, the sample was positioned between the light source and the integrating sphere. For the measurement of total reflectance, the sample was placed at the entrance of the integrating sphere, allowing the collection of radiation reflected by the tissue. For each sample, spectra were acquired in the spectral range between 200 and 1000 nm, which subsequently served as the basis for calculating the absorption spectra and for the spectral reconstruction described in the following sections.

For each tissue condition, 10 total transmittance spectra (T_t) and 10 total reflectance spectra (R_t) were acquired,

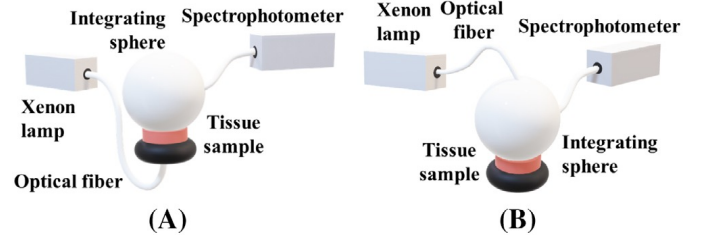


Fig. 2: Experimental setups used to measure the total transmittance (T_t) (A) and total reflectance (R_t) (B) spectra.

which served as the basis for determining the tissue absorption spectra. Additionally, reference absorption spectra for DNA, proteins, oxygenated hemoglobin (HbO_2), deoxygenated hemoglobin (Hb), water, lipids, melanin, and lipofuscin were considered as they represent some of the main absorbent components present in renal tissue and exhibit distinct spectral signatures across the analyzed wavelength range [14]. These reference spectra were used in the spectral reconstruction process to estimate the relative contribution of each absorbing component to the overall tissue spectrum.

C. Absorption Spectrum Calculation

Once the T_t and R_t spectra were obtained, the corresponding absorption coefficient spectra, $\mu_a(\lambda)$, were calculated using (1) presented in Section I. For the calculations, a sample thickness of $d = 0.05$ cm was considered, corresponding to the thickness of the samples used in the measurements. By applying (1), ten absorption spectra were obtained for healthy renal tissue and ten absorption spectra for tumor tissue (ChRCC). Subsequently, the individual spectra of $\mu_a(\lambda)$ were used to determine the average spectrum and the corresponding standard deviation for each tissue condition. The obtained average spectra allowed us to characterize the overall behavior of the optical absorption of the analyzed tissues, while the standard deviation provided information on the variability observed among the different samples. The average absorption spectra were subsequently used as an experimental reference in the spectral reconstruction process described in the following section.

D. Spectral Reconstruction

Prior to spectral reconstruction, the reference absorption spectra of the various tissue components considered in the study (DNA, proteins, oxygenated hemoglobin (HbO_2), deoxygenated hemoglobin (Hb), water, lipids, melanin, and lipofuscin) were normalized relative to their respective maximum absorption values. This procedure reduced the influence of amplitude differences between the various spectra, preserving only their relative spectral characteristics. Consequently, the comparison between the different components came to depend primarily on the shape of their absorption spectra, facilitating their use in the spectral reconstruction process.

After normalizing the reference spectra, they were organized to represent the different tissue components considered

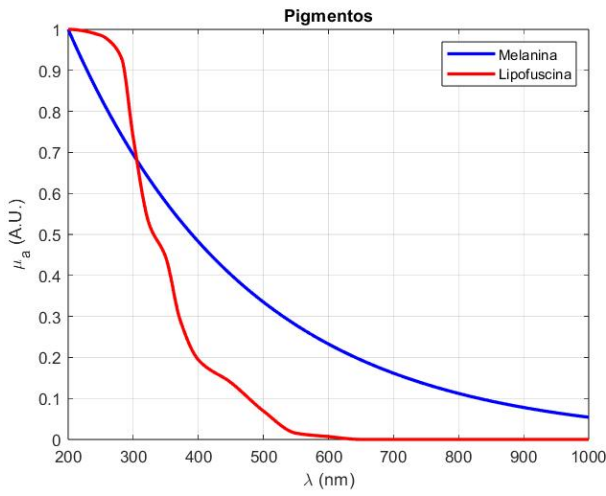


Fig. 3: $\mu_a(\lambda)$ spectra of melanin and lipofuscin between 200 and 1000 nm.

in the analysis. The spectra of DNA, proteins, oxygenated hemoglobin (HbO_2), deoxygenated hemoglobin (Hb), water, lipids, melanin, and lipofuscin were used as reference spectra for spectral reconstruction. In turn, the average absorption spectrum obtained for each tissue condition was used as an experimental reference in the fitting process. This approach enabled the comparison of the experimentally measured spectral behavior with the combination of the different reference spectra, allowing the relative contribution of each component to the overall tissue spectrum to be estimated according to the model described in (2).

The normalized absorption spectra of the tissue components considered in the reconstruction procedure are presented in Figures 3–5. These spectra were used as reference absorption profiles in the spectral reconstruction model.

As shown in Figures 3–5, the selected tissue components exhibit distinct absorption characteristics throughout the analyzed spectral range. DNA and proteins present higher absorption in the ultraviolet region, whereas oxygenated and deoxygenated hemoglobin exhibit characteristic absorption bands within the visible range [14]. Melanin and lipofuscin display broadband absorption behavior, while lipids present more pronounced absorption features at longer wavelengths. These differences in spectral behavior provide the basis for estimating the contribution of each component during spectral reconstruction.

To estimate the weights associated with each tissue component in (2), the non-negative least squares (NNLS) method was employed using the `lsqnonneg()` function in MATLAB. This method allowed us to determine the linear combination of the reference spectra that best reproduced the average experimental spectrum, minimizing the difference between the two while ensuring non-negative weighting coefficients for all components considered [20]. The weights obtained for each component were subsequently used to reconstruct the experimental spectrum through the weighted combination of

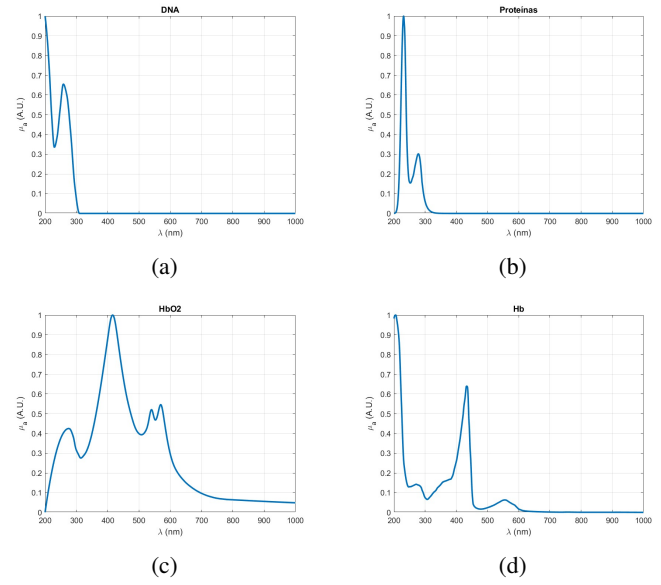


Fig. 4: $\mu_a(\lambda)$ spectra of DNA (A), proteins (B), oxygenated hemoglobin (C), and deoxygenated hemoglobin (D) between 200 and 1000 nm.

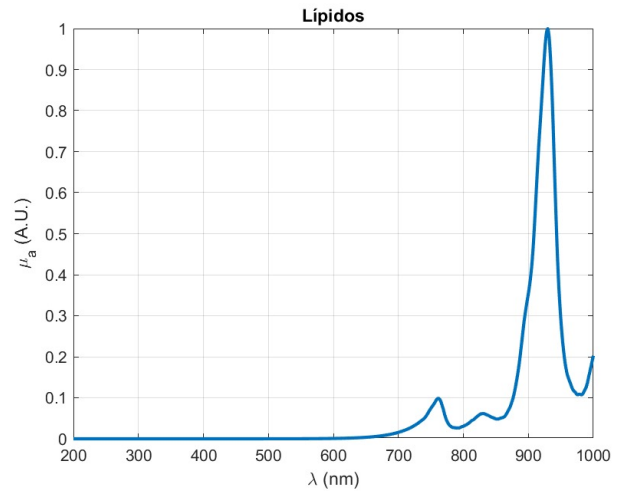


Fig. 5: $\mu_a(\lambda)$ spectrum of lipids between 200 and 1000 nm.

the respective reference spectra.

During the fitting phase, the contribution of lipids was found to be very small, with values close to zero. Since lipids are typically present in low concentrations in renal tissue and their spectral contribution is close to the sensitivity limit of the spectroscopic method, a slight adjustment of the fitting weights was performed to ensure that their contribution remained detectable without significantly affecting the overall quality of the spectral reconstruction. The resulting lipid contributions corresponded to approximately 0.01% for healthy tissue and 0.03% for tumor tissue (ChRCC), values that are consistent with previous reports in the literature [21], [22]. After this adjustment, the spectral fitting procedure was repeated using the NNLS algorithm.

E. Concentration Estimation

After spectral reconstruction, the final weights obtained for each tissue component were converted into relative concentrations. To this end, a volumetric water fraction of 77% was assumed, in accordance with values reported in the literature for healthy renal tissue [23]. The remaining 23% of the tissue volume was distributed among the other components considered in the analysis, namely DNA, proteins, oxygenated hemoglobin (HbO_2), deoxygenated hemoglobin (Hb), lipids, melanin, and lipofuscin. This distribution was performed proportionally to the weights obtained during the non-negative least squares fitting process. For each component, the final concentration was calculated based on its relative contribution to the spectral fit, ensuring that the sum of the concentrations of all constituents totaled 100%. The estimated concentrations for each component formed the basis of the comparative analysis between healthy and tumor tissues, presented in the following section.

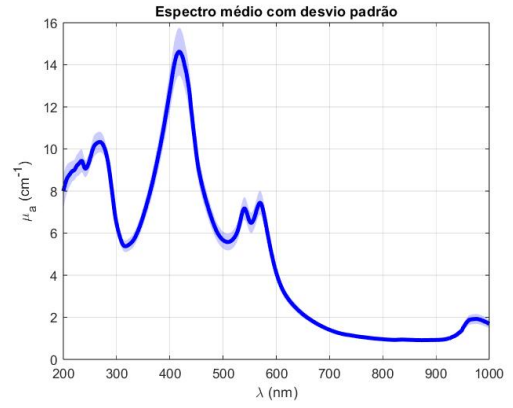
III. RESULTS AND DISCUSSION

The analysis began with the spectral measurements described in Section II. Using Equation (1), ten absorption coefficient spectra, $\mu_a(\lambda)$, were calculated for the healthy kidney samples and ten for the ChRCC samples from the corresponding total transmittance and total reflectance measurements. The resulting spectra covered the wavelength interval between 200 and 1000 nm.

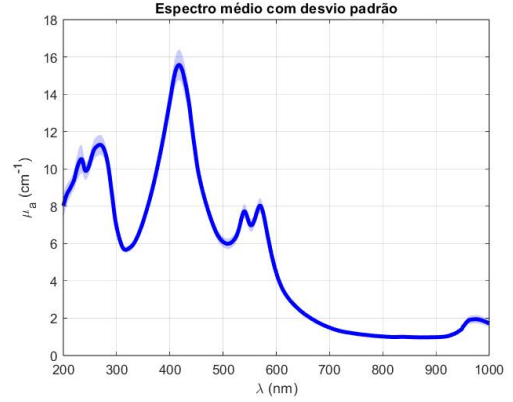
For each tissue condition, the mean $\mu_a(\lambda)$ spectrum and its corresponding standard deviation (SD) were determined from the individual measurements. These results are presented in Figure 6.

The spectra exhibit several characteristic absorption features associated with endogenous tissue chromophores [14]. In the ultraviolet region, an absorption band around 230 nm can be attributed to proteins, while the feature observed near 260–270 nm is related to DNA. Hemoglobin contributes the prominent Soret band centered at approximately 415 nm, together with the Q bands located near 540 and 577 nm. At longer wavelengths, a water absorption band is visible around 980 nm. In addition, the gradual decline of the absorption baseline throughout the visible range suggests the presence of pigments, particularly melanin and lipofuscin [14], [24].

Despite sharing the same overall spectral profile, healthy and ChRCC kidney tissues exhibit noticeable differences in absorption magnitude. For all major absorption bands, the ChRCC tissue shows slightly higher μ_a values. The Soret band reaches approximately 14.7 cm^{-1} in healthy tissue and 15.6 cm^{-1} in ChRCC tissue, while the UV absorption peak near 270 nm increases from about 10 cm^{-1} to 11 cm^{-1} . Similarly, the hemoglobin Q bands around 540–577 nm are somewhat stronger in ChRCC tissue ($\sim 8.1 \text{ cm}^{-1}$ compared with $\sim 7.7 \text{ cm}^{-1}$). In contrast, the water absorption band near 980 nm remains essentially unchanged at approximately 2.0 cm^{-1} for both conditions. These observations suggest increased concentrations of proteins and DNA in the ChRCC



(a) Healthy kidney tissue.



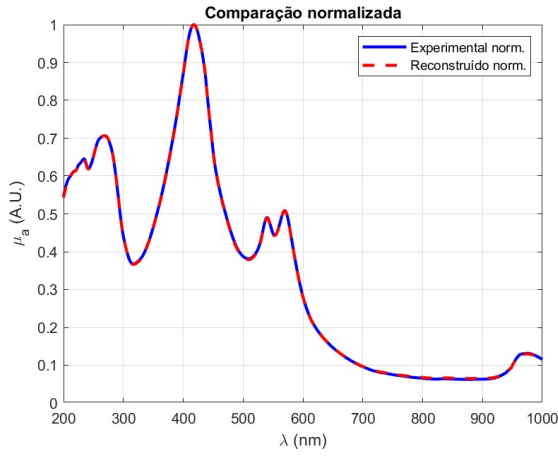
(b) ChRCC kidney tissue.

Fig. 6: Mean absorption coefficient spectra (μ_a) and corresponding standard deviations for healthy and ChRCC kidney tissues.

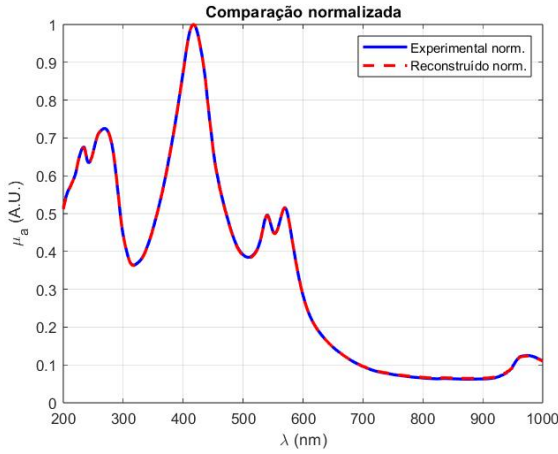
samples relative to healthy tissue, together with subtle changes in blood chromophores.

To obtain further quantitative information, spectral reconstruction of the $\mu_a(\lambda)$ curves was performed according to the procedure outlined in Section II. The optimal chromophore weighting factors (w_i) were determined through a least squares fitting approach. Figure 7 presents the normalized comparison between the experimental and reconstructed mean absorption spectra for both healthy and ChRCC kidney tissues.

The agreement between measured and reconstructed spectra is excellent for both tissue types. Across the entire spectral range, the reconstructed curves reproduce the experimental profiles and accurately capture all relevant absorption features. Furthermore, the reconstructed spectra remain within the SD limits of the experimental averages for almost all wavelengths. A small deviation can be observed in the vicinity of the water absorption band around 980 nm, where the reconstructed profile appears slightly broader. This discrepancy arises from the use of the absorption spectrum of pure liquid water during reconstruction. In biological tissues, a substantial fraction of water is bound to macromolecular structures, which results in a narrower absorption band than that of pure water [14].



(a) Healthy kidney tissue.



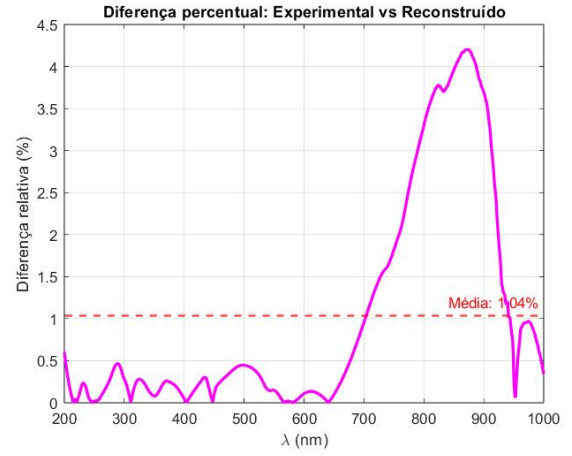
(b) ChRCC kidney tissue.

Fig. 7: Normalized comparison between the experimental and reconstructed absorption spectra for healthy and ChRCC kidney tissues.

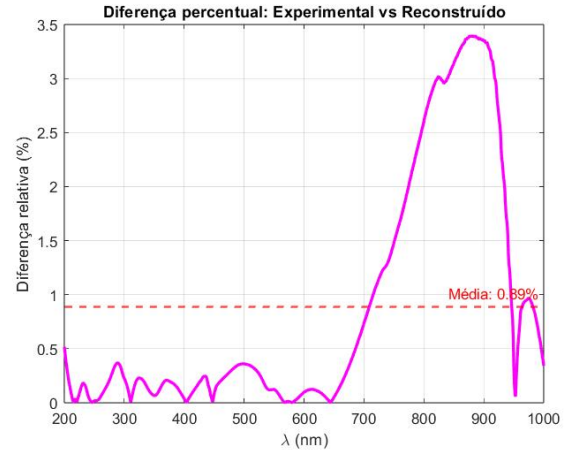
Given that kidney tissue contains approximately 77% water by volume, with a significant portion existing in bound states, this behavior is expected and does not significantly affect the quality of the reconstruction [14].

The accuracy of the reconstruction was further evaluated by calculating the relative percentage difference at each wavelength between the experimental and reconstructed spectra. The results are presented in Figure 8 for both healthy and ChRCC kidney tissues.

Mean relative differences of 1.04% and 0.89% were obtained for healthy and ChRCC tissues, respectively, indicating a high level of agreement. Within the ultraviolet and visible spectral regions (200–600 nm), which contain the most diagnostically relevant absorption features, the reconstruction error remains below 0.5% in both cases. The largest discrepancies occur in the near infrared region between approximately 800 and 950 nm, reaching values of about 4.2% for healthy tissue and 3.4% for ChRCC tissue. These deviations are consistent with the previously discussed mismatch associated with the



(a) Healthy kidney tissue.



(b) ChRCC kidney tissue.

Fig. 8: Relative percentage difference between the experimental and reconstructed absorption spectra for healthy and ChRCC kidney tissues.

water absorption band.

The weighting coefficients obtained from the least squares fitting procedure were subsequently used to estimate chromophore concentrations. Assuming a total water volume fraction of 77% in both tissue conditions, a reasonable assumption considering that cancer progression may convert bound water into mobile water without substantially changing the overall water content, the concentrations of the remaining chromophores were calculated. The complete set of weighting factors and derived concentrations is reported in Table I.

The concentration values presented in Table I reveal differences between healthy and ChRCC kidney tissues. DNA concentration increases from 3.05% in healthy tissue to 3.28% in ChRCC tissue, while protein concentration rises from 1.66% to 2.17%. These increases are consistent with the enhanced cellular proliferation and metabolic activity commonly associated with cancer development [6], [25], [26]. Lipids remain present at low levels in both tissue types but exhibit a moderate increase from 0.01% to 0.03% in the ChRCC samples.

TABLE I: Chromophore weights (p_i) and concentrations obtained from spectral reconstruction.

Kidney	Chromophore	p_i	Conc. (%)
Healthy	DNA	2.9892	3.0536
	Proteins	1.6225	1.6575
	HbO ₂	12.9240	13.2027
	Hb	2.3486	2.3992
	Water	1.1948	77.0000
	Lipids	0.0103	0.0105
	Melanin	0.9120	0.9317
	Lipofuscin	1.7079	1.7447
ChRCC	DNA	3.4908	3.2827
	Proteins	2.3063	2.1688
	HbO ₂	14.1371	13.2943
	Hb	1.8656	1.7544
	Water	1.1653	77.0000
	Lipids	0.0334	0.0314
	Melanin	0.7223	0.6792
	Lipofuscin	1.9026	1.7892

Although no distinct lipid absorption feature can be readily identified in the experimental spectra shown in Figure 6, the reconstruction procedure was sufficiently sensitive to detect their contribution in both tissue conditions.

Regarding the blood chromophores, HbO₂ increases slightly from 13.20% to 13.29%, whereas Hb decreases from 2.40% to 1.75%. The pigment concentrations also exhibit distinct changes. Melanin decreases from 0.93% to 0.68%, whereas lipofuscin shows a slight increase from 1.74% to 1.79%. Although these variations are relatively small, they indicate alterations in tissue composition associated with the tumor state [25], [26].

Overall, the most pronounced differences between healthy and ChRCC tissues are observed for DNA and protein content, suggesting that these chromophores may provide relevant diagnostic contrast for distinguishing between the two conditions.

Taken together, the chromophore concentrations obtained from spectral reconstruction reveal measurable biochemical differences between healthy and ChRCC kidney tissues. The observed variations are consistent with established molecular and physiological changes associated with ChRCC [6], [26]. These findings demonstrate the potential of broadband absorption spectroscopy combined with least squares spectral reconstruction as a promising basis for future minimally invasive diagnostic methodologies.

IV. CONCLUSIONS

In this work, the absorption spectra of healthy renal tissue and ChRCC were successfully reconstructed using the absorption spectra of their principal biological components and a non-negative least squares fitting approach. The reconstructed spectra showed excellent agreement with the experimental data across the entire spectral range, with mean relative differences below 1.1% for both tissue conditions, demonstrating the robustness and accuracy of the proposed methodology.

The spectral analysis revealed measurable differences between healthy and tumour tissues, particularly in the estimated concentrations of DNA and proteins, which were found to be higher in ChRCC. These findings are consistent with the

increased cellular proliferation and metabolic activity associated with tumour development. Additional variations were observed in the concentrations of haemoglobin derivatives, lipids, melanin, and lipofuscin, reflecting biochemical alterations associated with the cancerous state.

In general, the results demonstrate that broadband absorption spectroscopy combined with non-negative least squares spectral reconstruction can provide quantitative information regarding tissue composition and chromophore content. This approach represents a promising tool for tissue characterization and may contribute to the development of future minimally invasive optical diagnostic techniques for the identification and discrimination of renal tumours.

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